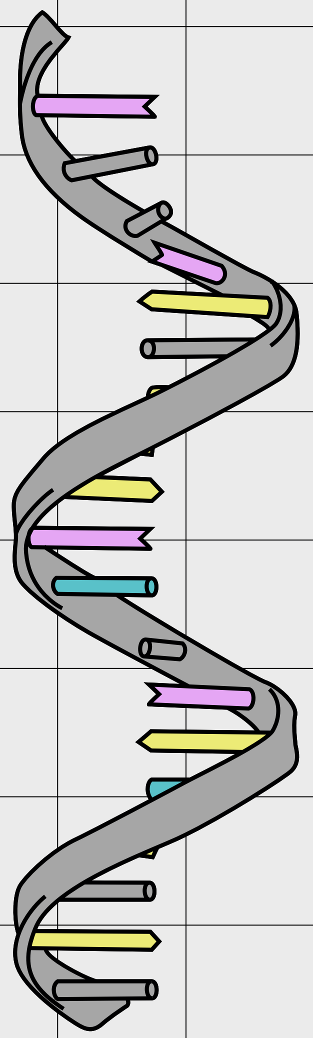
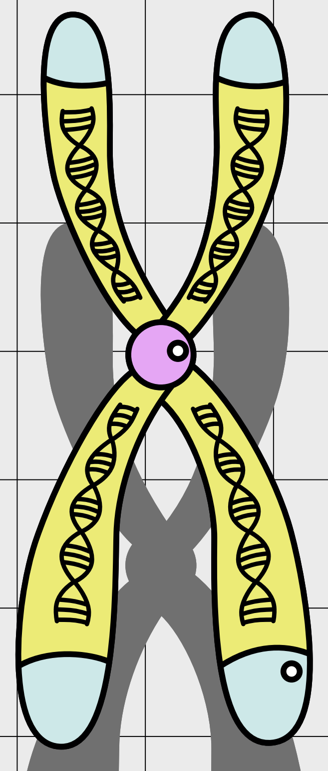


MULTISCALE MODELLING OF IMMUNOGENIC CELL DEATH

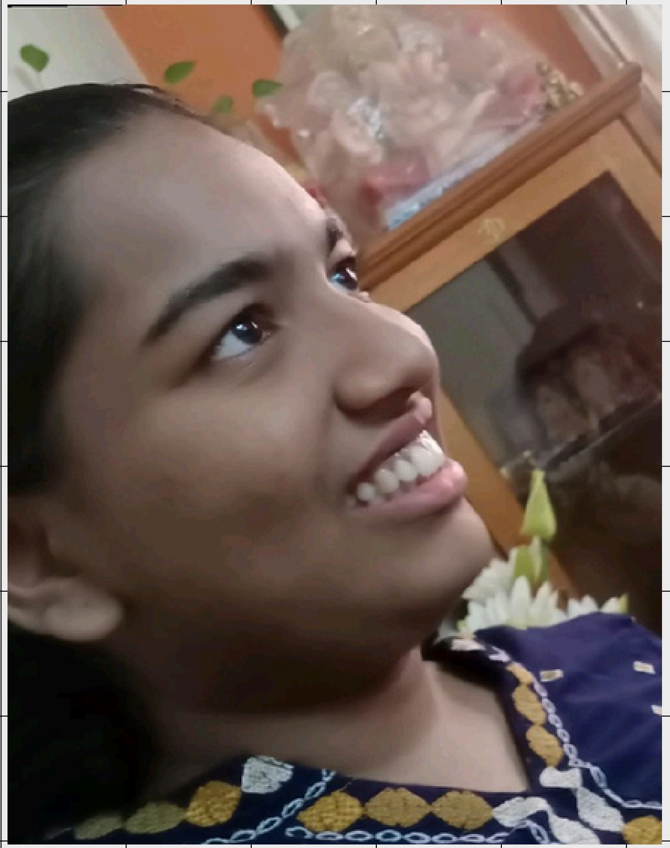
Project ID: BIL24P01
Mentor: A Arun Kumar
PI: Dr Jhinuk Chatterjee



Department of Biotechnology
PES University, Bengaluru-560085

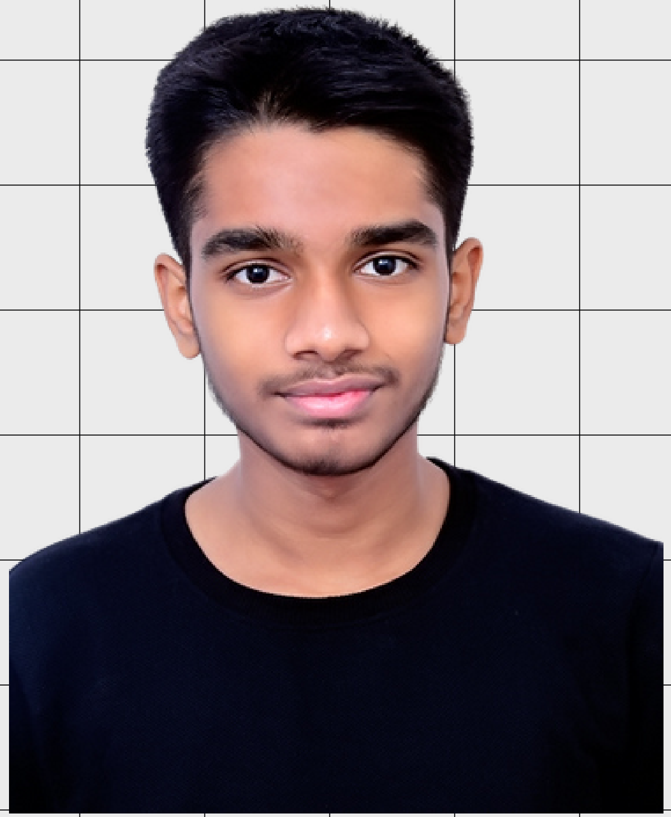
TABLE OF CONTENTS

OUR TEAM



Pragathi Prasad, 5th Sem BT

Sanjana Rao, 5th Sem BT



Rahul Jaikrishna, 3rd Sem CS

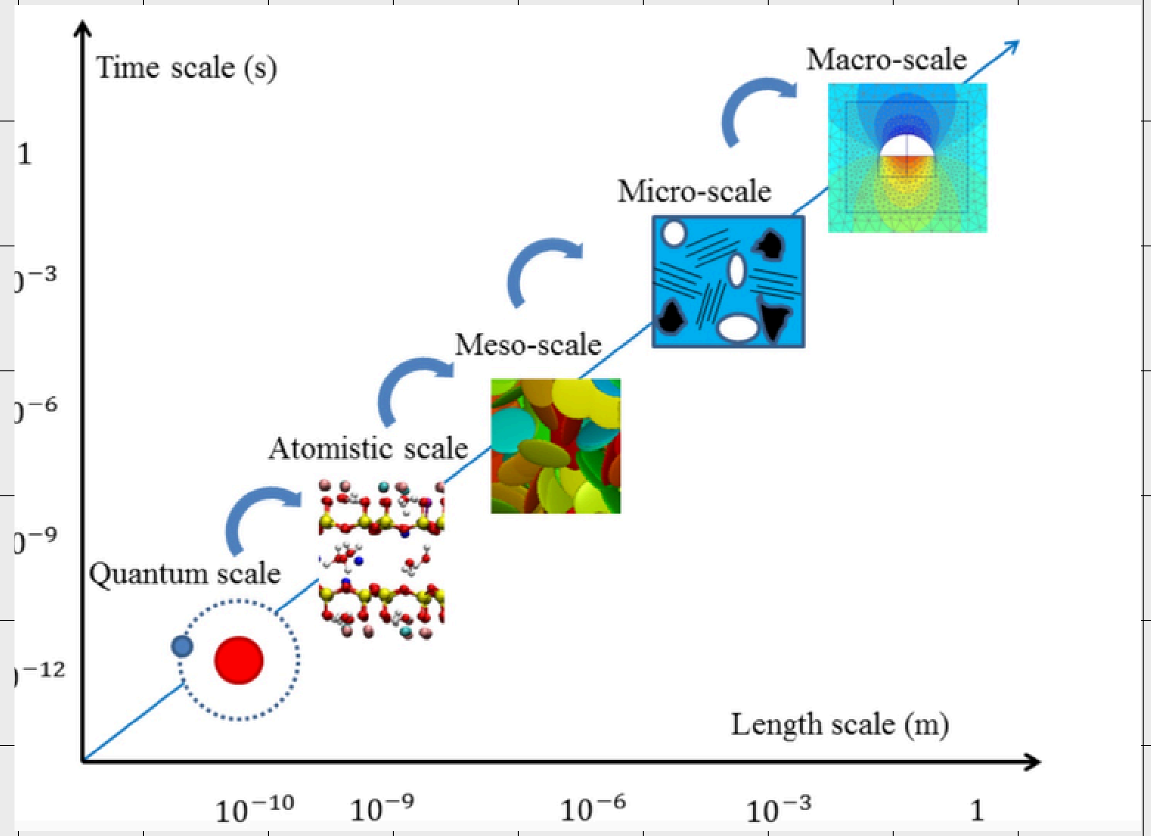
Shobita Shankaran, 3rd Sem BT

INTRODUCTION

Cancer is a malignant and invasive disease which has been developing resistance to chemotherapy and drugs alike. There is a need to develop methods to counteract this problem. While several wet and dry lab techniques have been developed for the same, observing interactions between elements is quite a challenge seen using wet lab methods, , in-silico programs using the concepts systems biology have proven to be a gold standard in being able to observe complex interactions at a cellular level

Multi-scale modeling, as the name suggests, is a way of linking biological phenomenon of different scales together. It links the interacting molecular species to the larger picture of how cancer manifest in the body, by linking the logic of the pathway to how the cells behave. We have solved this problem in 2 main stages :

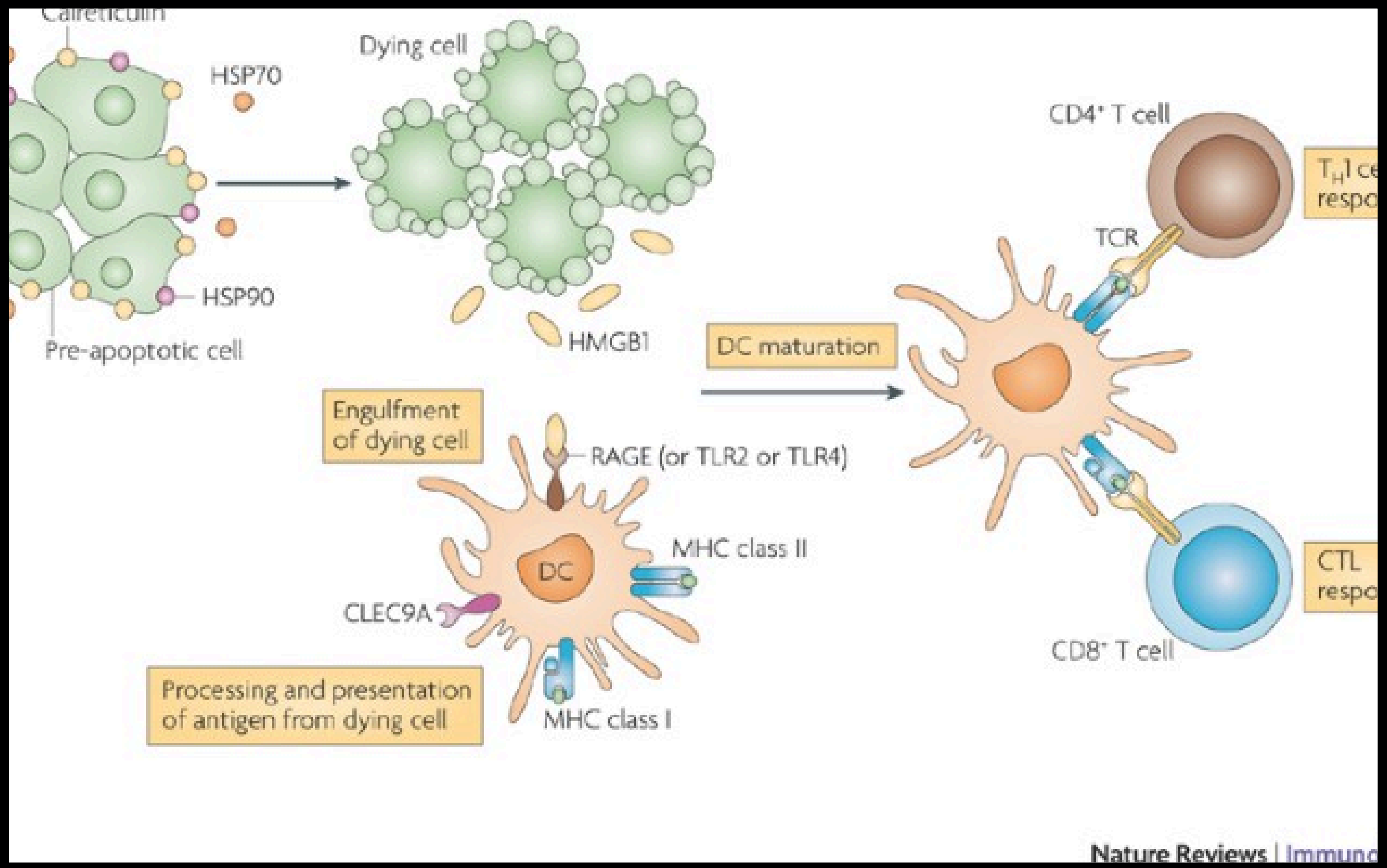
- Modelled the pathway using Boolean Networks
- Modelling the cells using Agent Based Modelling

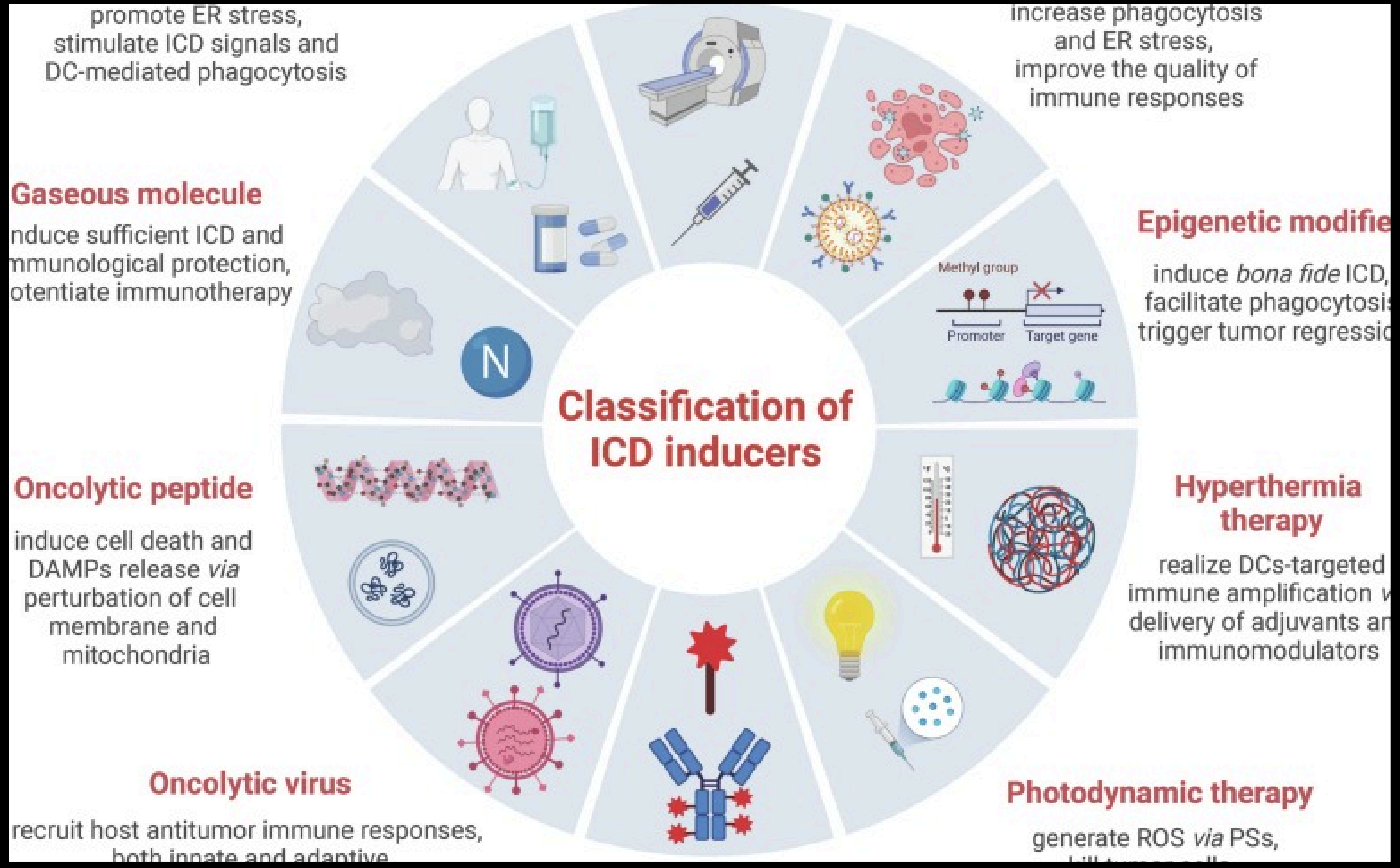


https://ajw-group.mit.edu/multiscale-modeling-clays

IMMUNOGENIC CELL DEATH

- Regulated cell death
- Activates an adaptive immune response
- ICD of cancer cells release signals (DAMPs) that activate APCs
- APC'S present antigens to T cells
- Can be induced by chemotherapy, radiotherapy, oncolytic viruses, antibody therapy, etc.

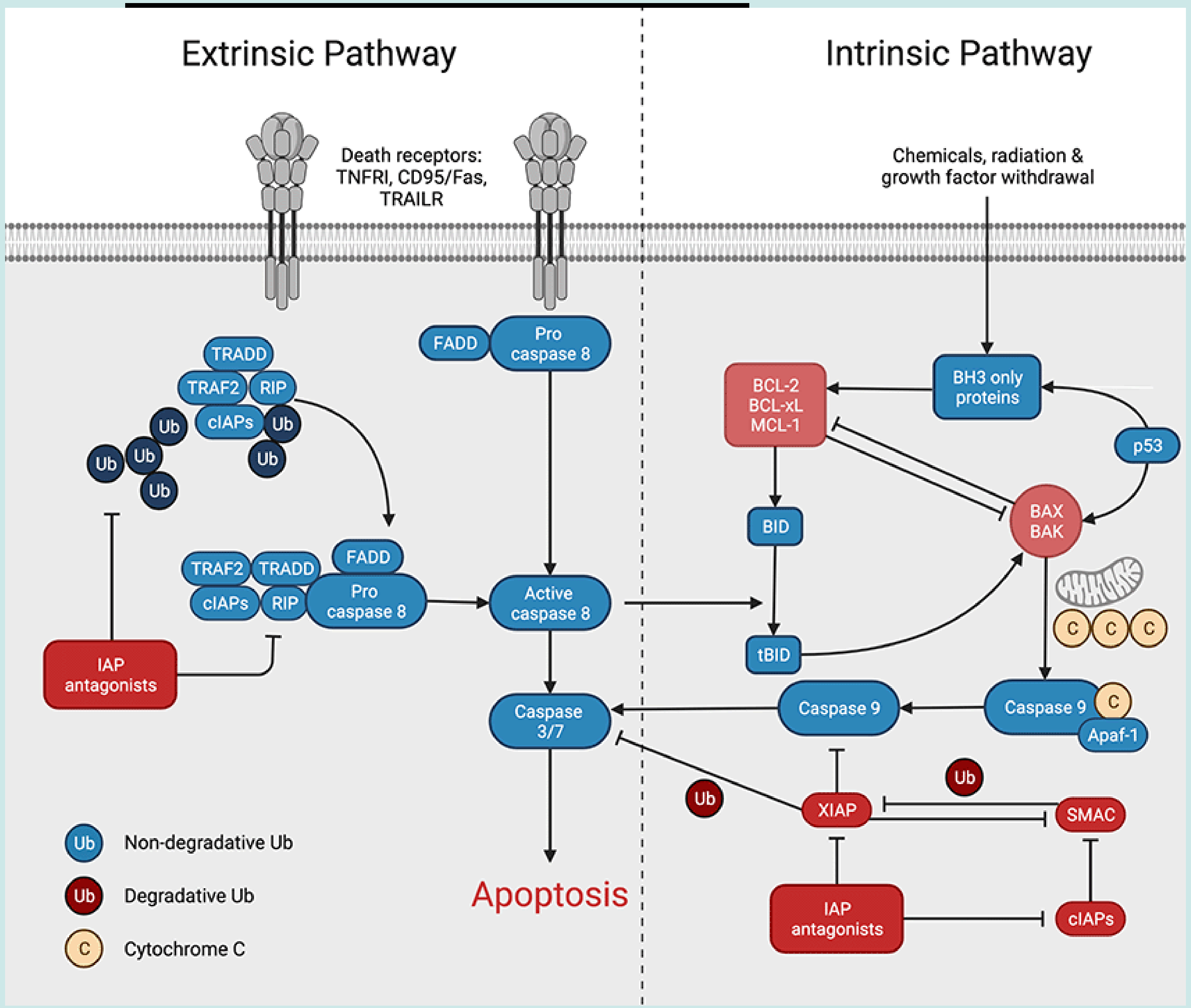




source: https://www.nature.com/articles/nri2545

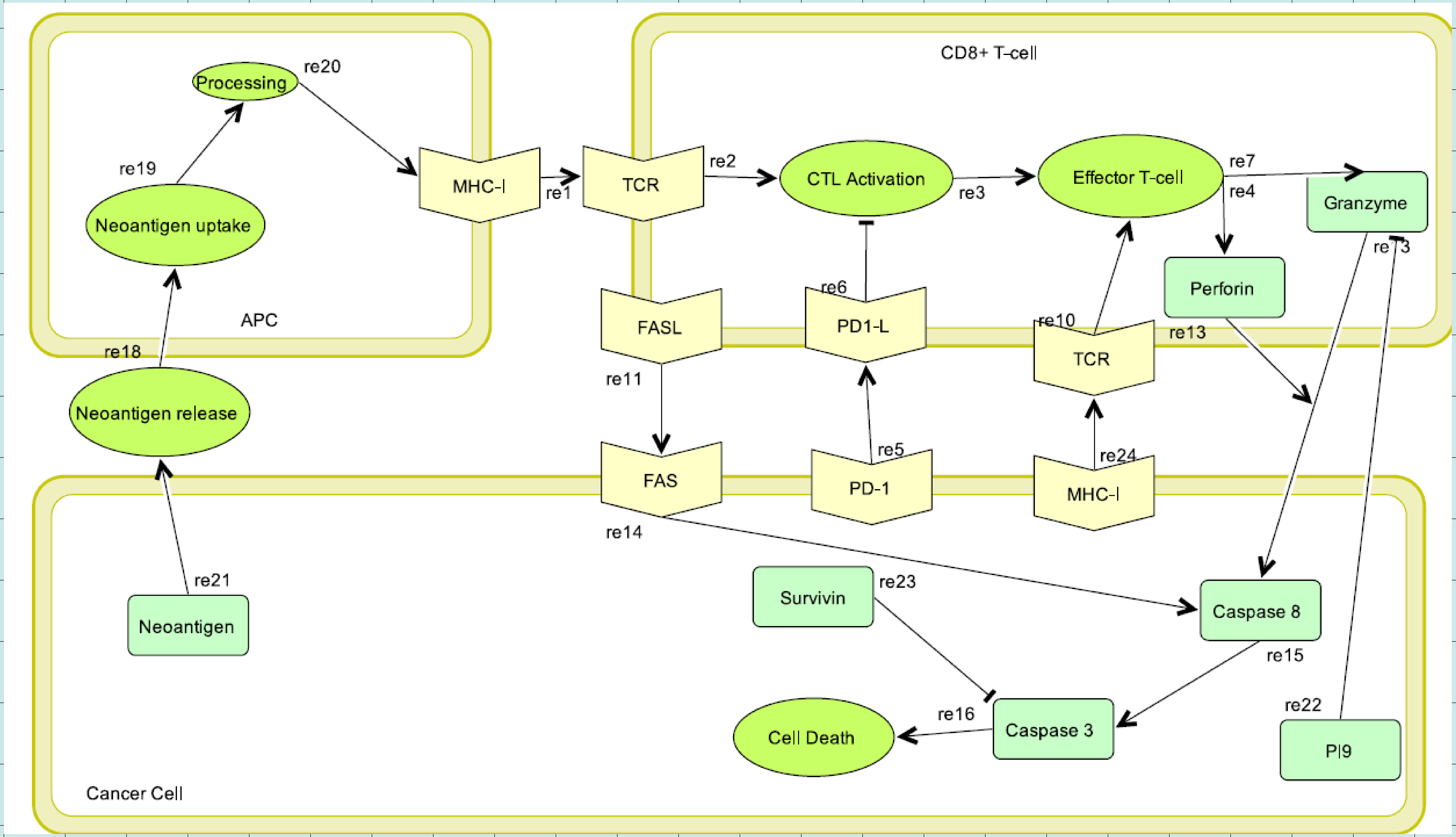
PROGRAMMED CELL DEATH

- Controls elimination of cells
- Maintains cellular and organismal homeostasis.
- Can be triggered via 2 pathways:
- **Extrinsic Pathway:** binding of extracellular death ligands to death receptors
- **Intrinsic Pathway:** triggered by internal cellular stress signals



source:https://www.rockland.com

PATHWAY DESIGNED



TOOLS USED

WebMaBoSS

WebMaBoSS is an online tool for simulating Boolean models of biological networks, allowing users to explore how different gene or protein interactions influence cellular behavior

PhysiCell & PhysiCell studio

PhysiCell was used to model intercellular interactions within the tumor microenvironment, simulating how cancer cells interact with surrounding cells and the extracellular matrix.

METHODOLOGY

The two main elelments of the experimental methodology include pathway simulation and graphical tumor simulation

PATHWAY SIMULATION

- .xml file converted into a qualitative SBML file with the help of a Python module CasQ
- SBML qual file was loaded into WebMaBOSS,to retrieve boolean relationship between elements
- The initial values and output conditions were set the the simulation was done for 10000 cycles, for 100 seconds, with 1 second time tick and BND and CFG files were generated

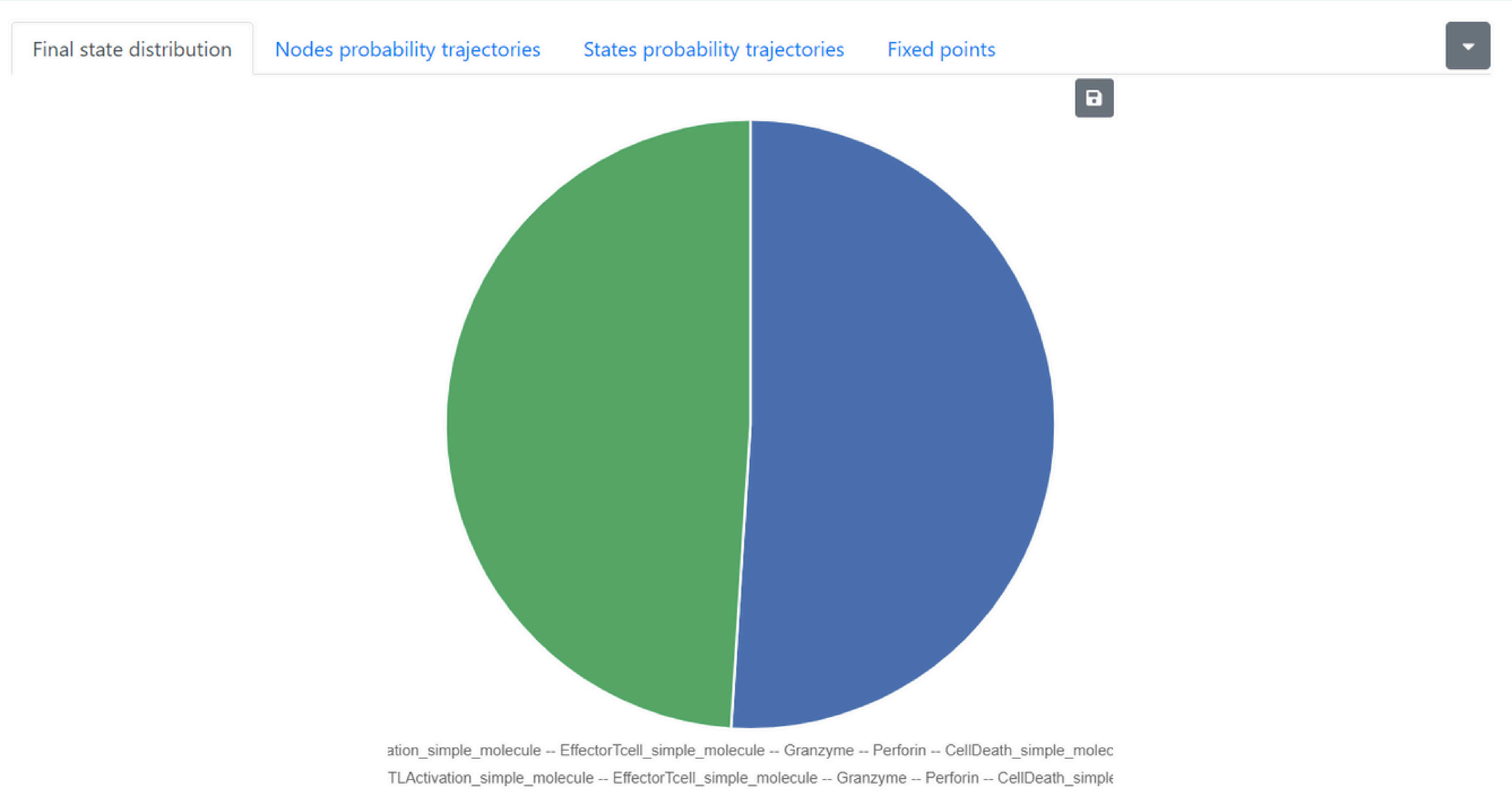
TUMOR SIMULATION

- The .bnd and .cfg files were loaded onto to PhysiCell studio and cell types were defined
- Other initial conditions like motility, microenvironment, rules, etc were set, while some conditions were set as default
- The .exe file of the project was complied priority and its path was set before the simulation was started.
- The simulation ran for 2hs, recorded for a 24h time frame. A snapshot was taken for every 2s interval and the output .svg and .jpg files were stored in the output folder
- The generated .jpg files were combined together using ffmpeg in proper order, with each frame timed at 0.05s. The final output video was downloaded.

RESULTS

WebMaBOSS

- Two final state distribution posibilities are obtain almost equally: one involving PD1L and PD1 and the other not involving both

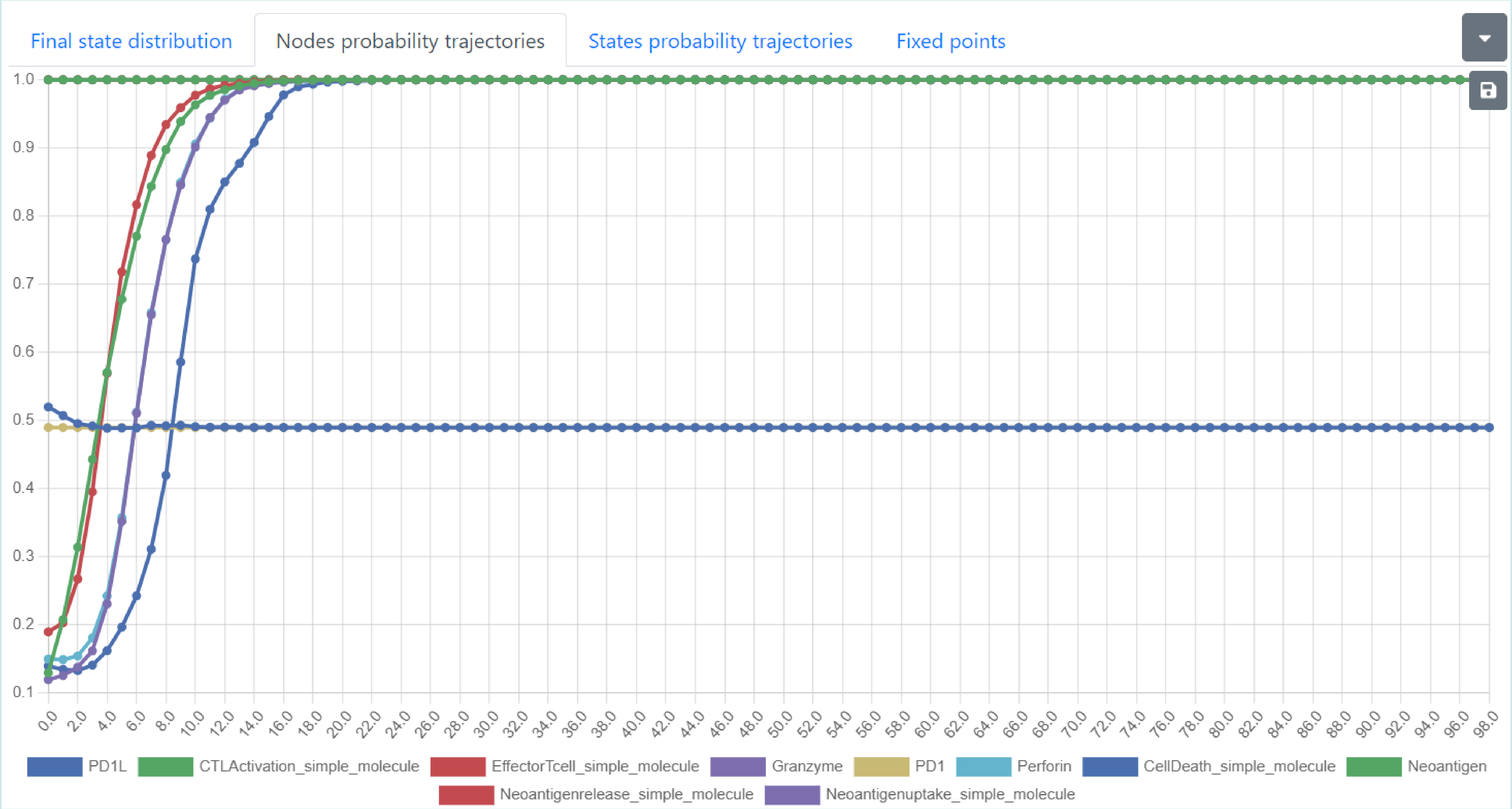


Final state distribution graph

RESULTS

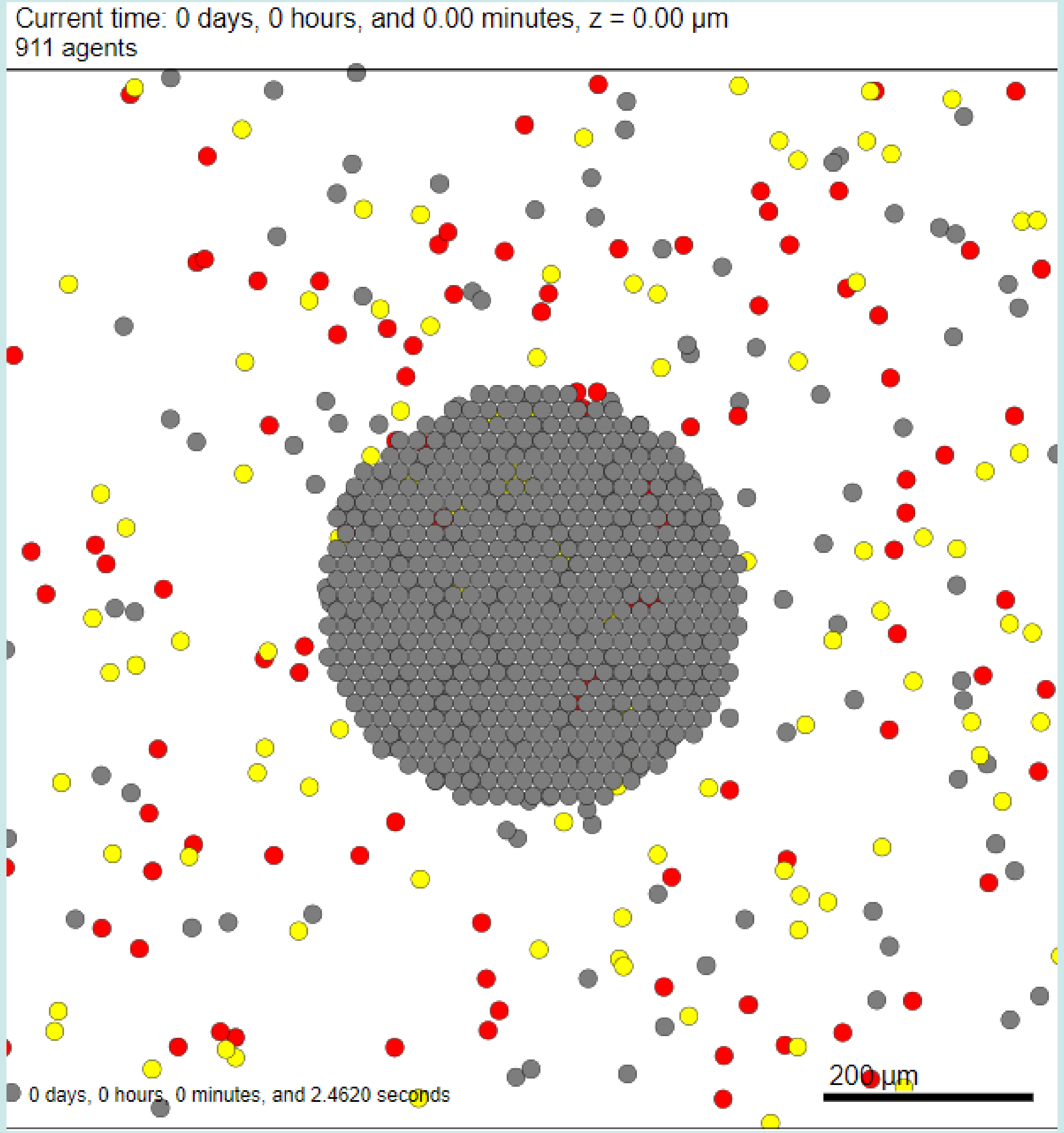
WebMaBoSS:

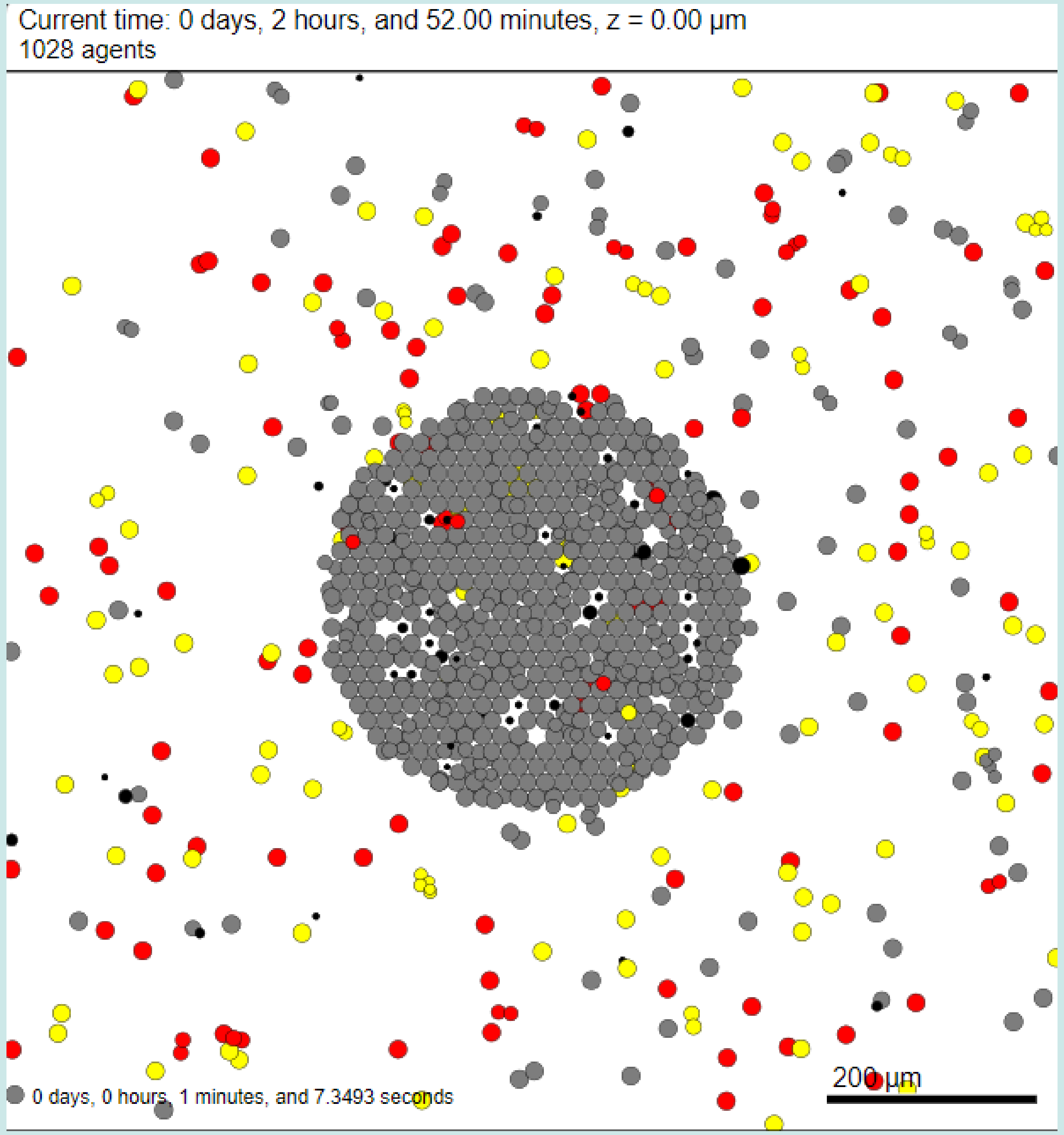
- As time passes the node probability trajectories of Cell death, Granenzyme, Perforin, Effector T-cell and CTL activation molecules increase in a sigmoid fashion

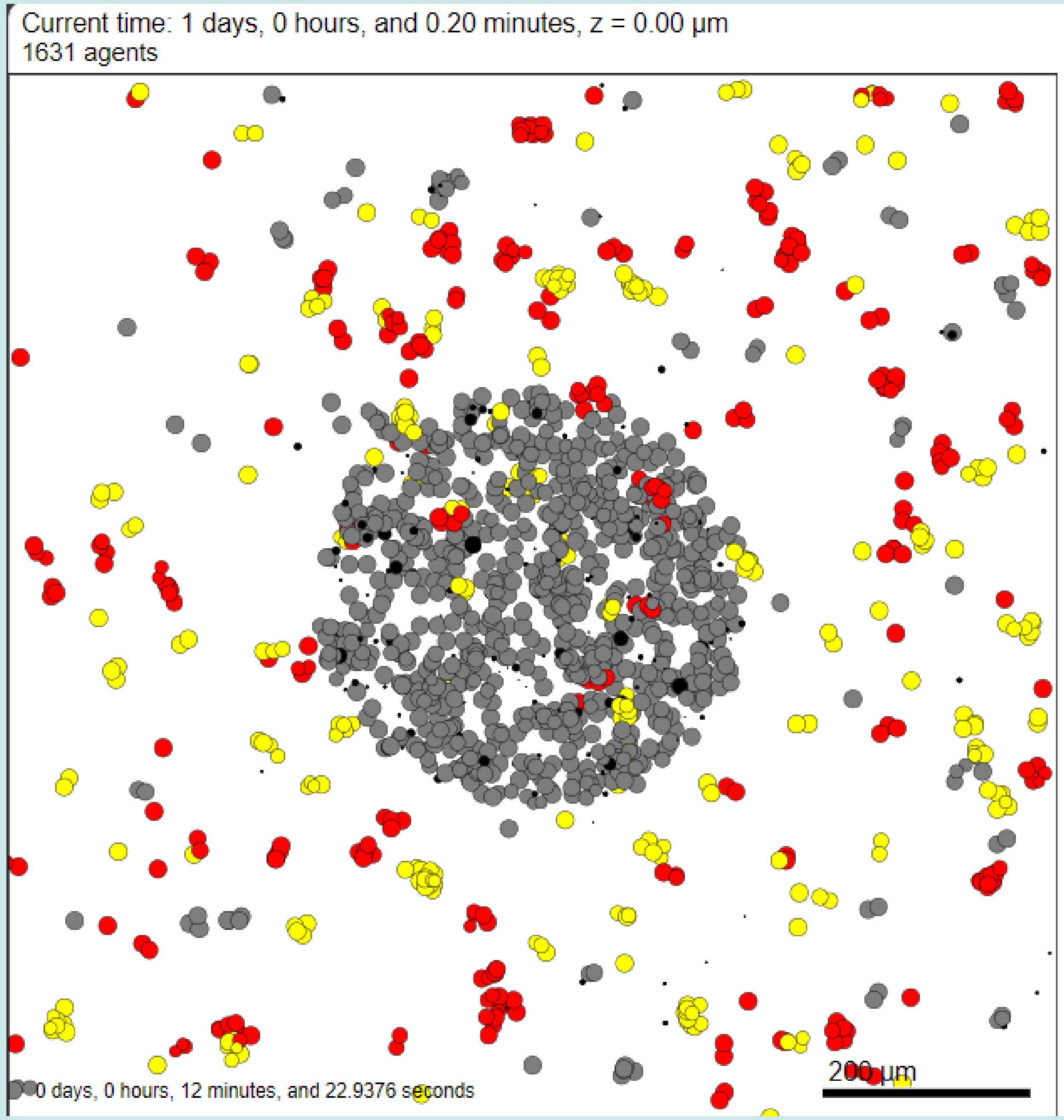


Node probability trajectory graph

RESULTS - PHYSICELL



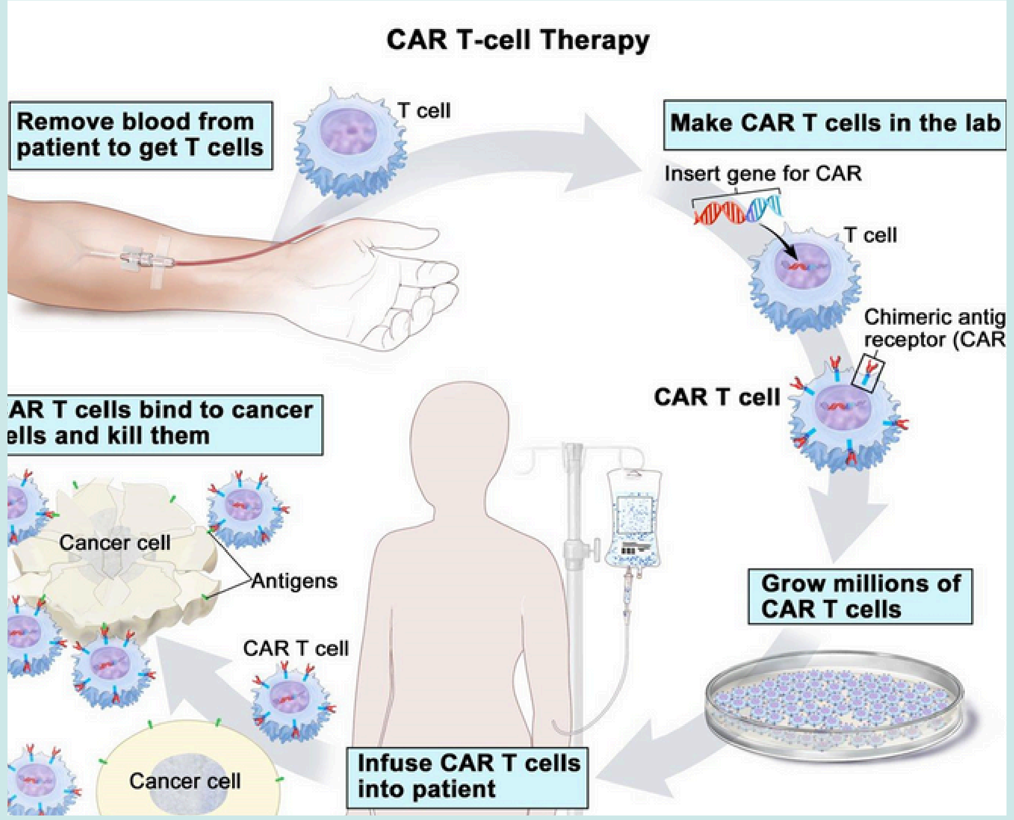




LIMITATIONS

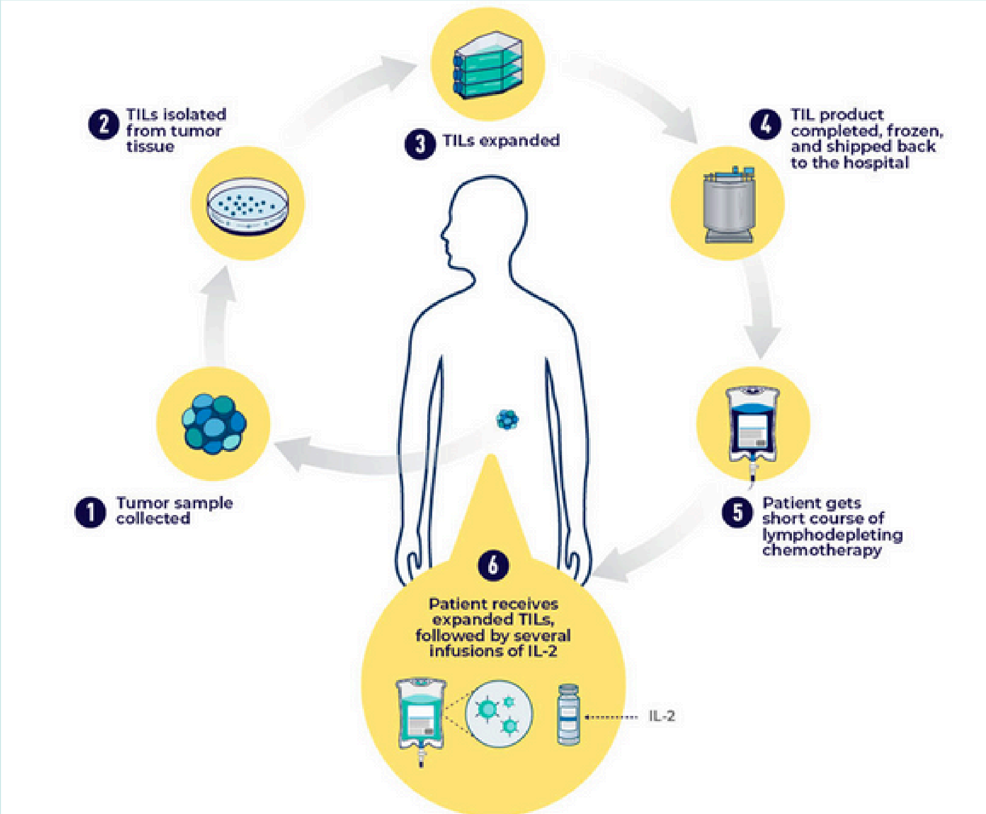
- We could includ only 2 caspases in our pathway even though multiple are shown to interact
- We have not included parameter optimization , most of the values are set to default.
- Abnormal reactions were seen where helper cells were dying, a phenomenon we couldn’t quite describe.
- The mathematical basis were not fully described due to knowledge gaps

FUTURE SCOPE



CAR T CELL THERAPY

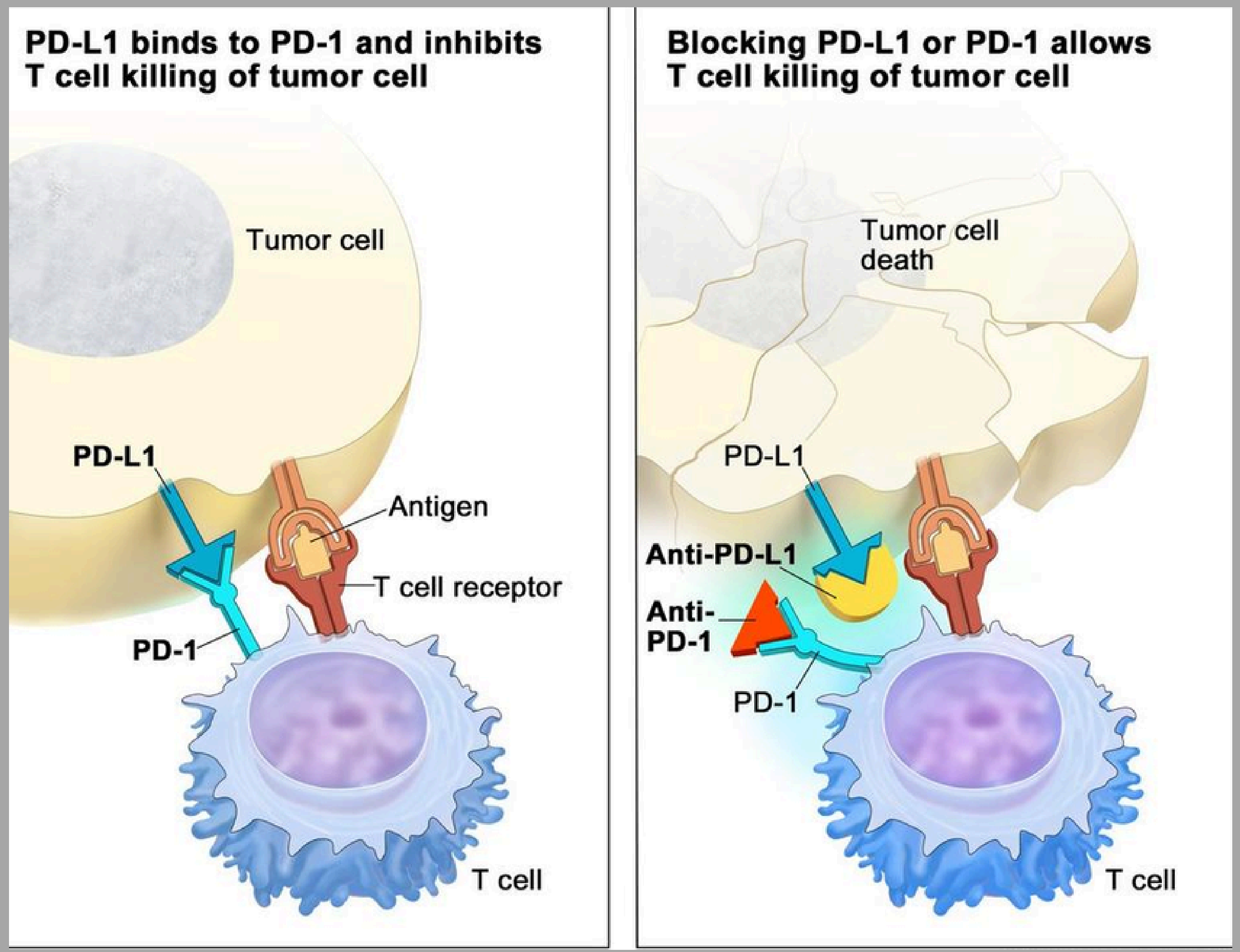
- Developing a larger , more inclusive pathway
- Completing cancer cell modelling using a compartmental modelling approach (showing how cancer cells migrate from one part of the body to another)
- Inculcation of various treatment methods (CAR T cell therapy , t cell transfer therapy, immune checkpoint inhibition etc) into the model and simulating their effects on the tumor



T CELL TRANSFER THERAPY

IMMUNE CHECKPOINT INHIBITION

- Prevent an autoimmune response from occurring
- Proteins on t cells bind to immune checkpoint proteins
- This binding send an “OFF” signal to the t cells
- ICI’s block this binding from happening
- ICI drugs : Ipilimumab ,Nivolumab etc.



THANK YOU!